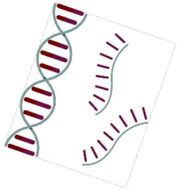
RNA and DNA library prep for mNGS

mNGS pipeline

Study design

- Type(s)
- Controls
- Storage
- Lab setup

xNA extraction



- Type(s)
- Controls
- QC

Library prep

- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

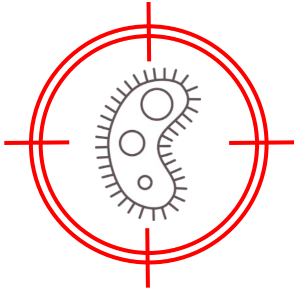
Sequencing

- Machine
- Read length
- Loading conc.

Data analysis

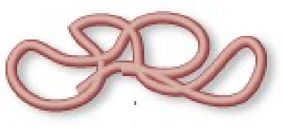
- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

Pathogen ID

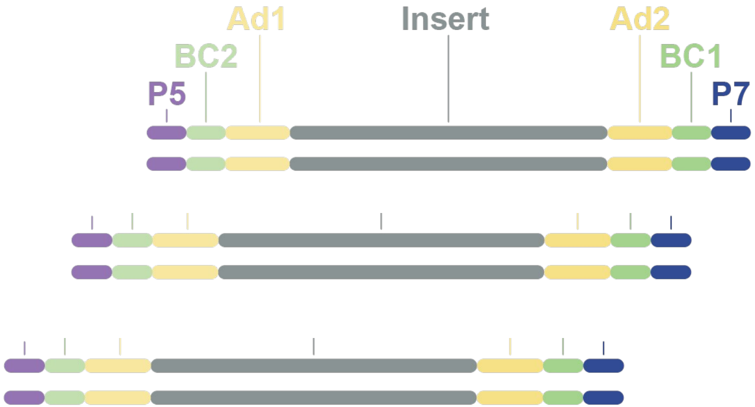


Library prep: RNA to sequencing library

- Sequencing library = Pool of DNA inserts with sequencing adapters attached



Extracted RNA



Sequencing library

Steps in library preparation:

1. Host depletion, RNA fragmentation + priming

2. First strand cDNA synthesis

3. Second strand cDNA synthesis

4. End Prep

5. Adapter Ligation

6.. USER + Barcoding PCR

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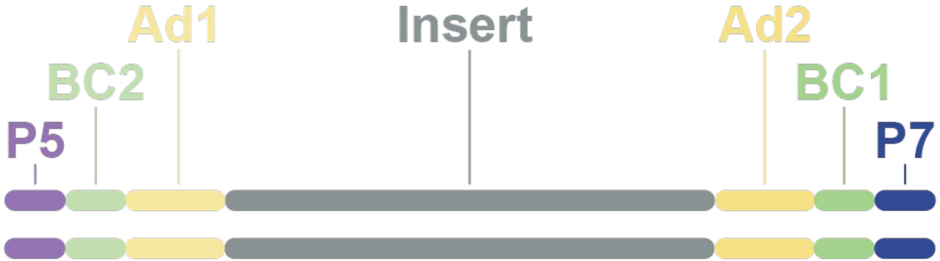
Library prep

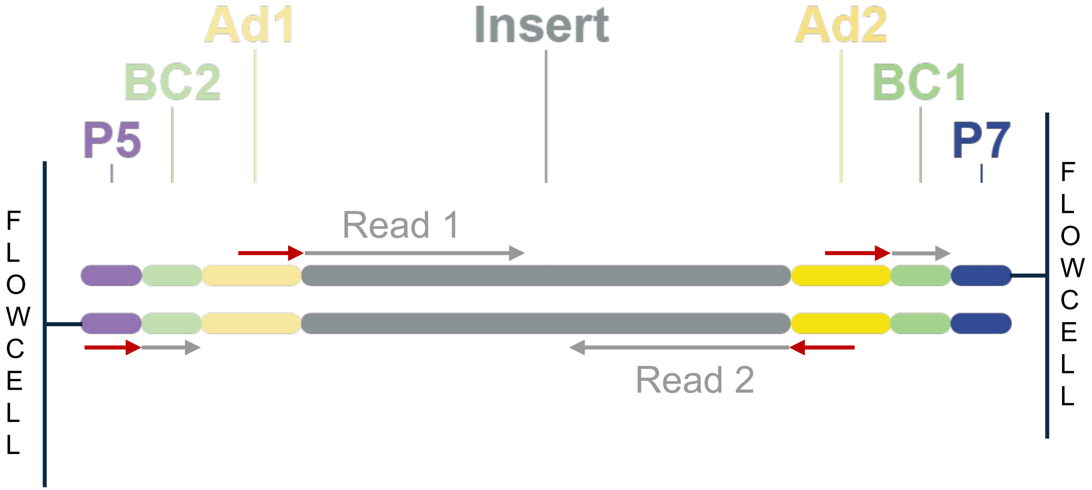
- Sequencing library = Pool of DNA inserts with sequencing adapters attached



Insert	= DNA fragment to be sequenced
Ad1/Ad2	= Adapters that allow initiation of sequencing
BC1/BC2	= Barcodes specific for each sample (one for each strand)
P5/P7	= Flow cell binding sites

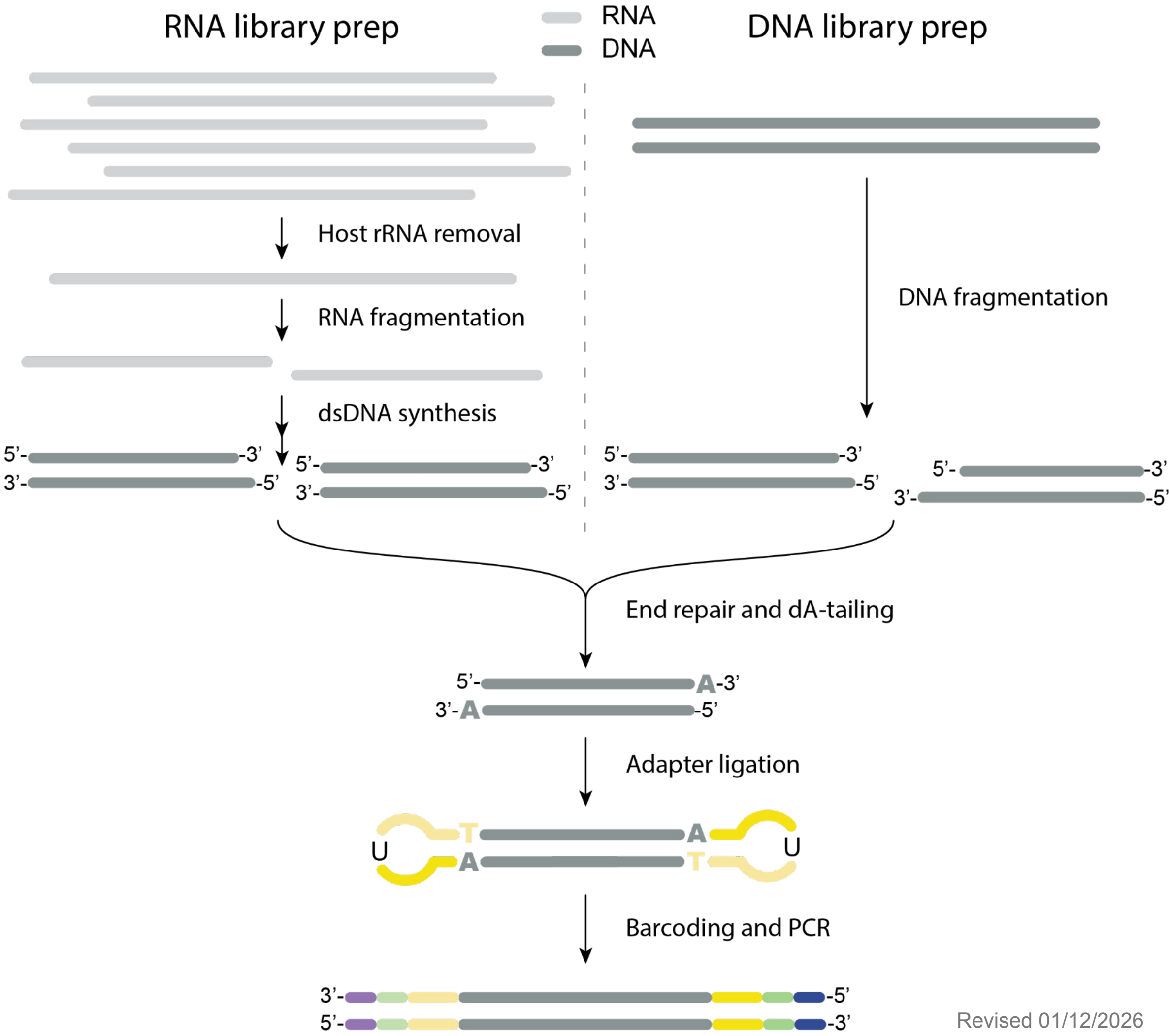
Library prep

- Sequencing process:



Library prep

- RNA or DNA library



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RNA library prep

STEP 0. Starting material: RNA or TNA

- RNA only: move on to host rRNA depletion
- TNA (total nucleic acid): need to remove DNA
    - Aliquot sample
    - DNAse I treatment
    - SPRI bead clean-up (2x)
    - Qubit quantification

RNA library prep

RNA content of typical host sample
95% host rRNA
2-3% host mRNA
1% other host RNA
<1% pathogen RNA

STEP 1. Host rRNA depletion

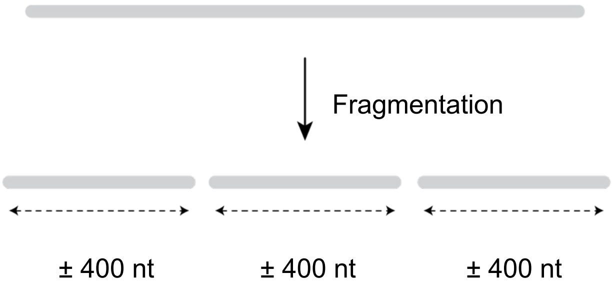
- Before RNA extraction:
  - Serum or plasma instead of blood
  - Supernatant of CSF to remove WBCs

- Human RNA depletion kits to use after RNA extraction:
  - TruSeq Stranded total RNA Human (previously Ribo-Zero, Illumina)
    - Biotinylated, target-specific oligos + Ribo-Zero beads to fish out rRNA
  - FastSelect (Qiagen): $150 for 24 samples when reagent is diluted 1:10
    - Proprietary technology, presumably with hybridizing probes that make rRNA inaccessible to enzymes in subsequent steps

Preferred
(no extra wash required)

RNA library prep

STEP 2. Fragmentation

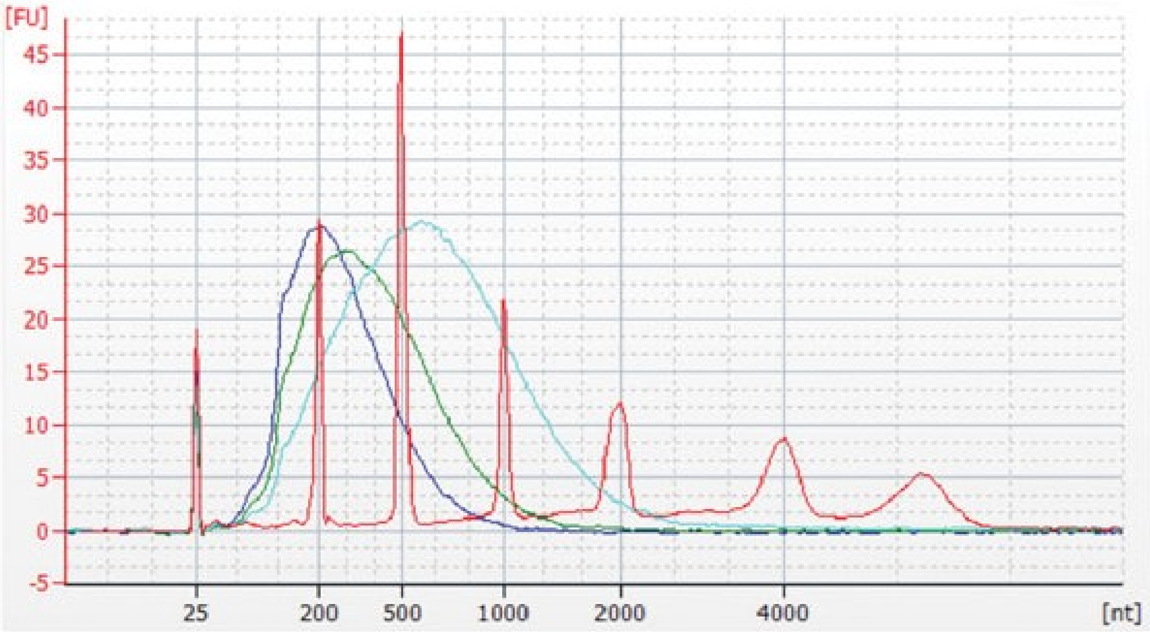


- Ideal insert size = ± 400 nt
- Several ways to fragment RNA:
  a. Enzymatic: %GC-dependent -> may bias mNGS results
  b. Heat cycling: relatively random
  c. Sonication: larger volume required, expensive machine, contamination risk
  d. Divalent cations: allow RNA to be self-cleaving (Mg^{2+}, Ca^{2+}, Zn^{2+}, …)
     - Mg^{2+} and heat (94°C) for 2-15 minutes depending on RNA quality
       - High quality RNA: long fragmentation time
       - Low quality RNA: short fragmentation time

NEBNext Ultra ii RNA library kit

RNA library prep

Fragmentation pattern of high quality RNA



STEP 2. Fragmentation

Use shorter fragmentation times when RNA quality is compromised!

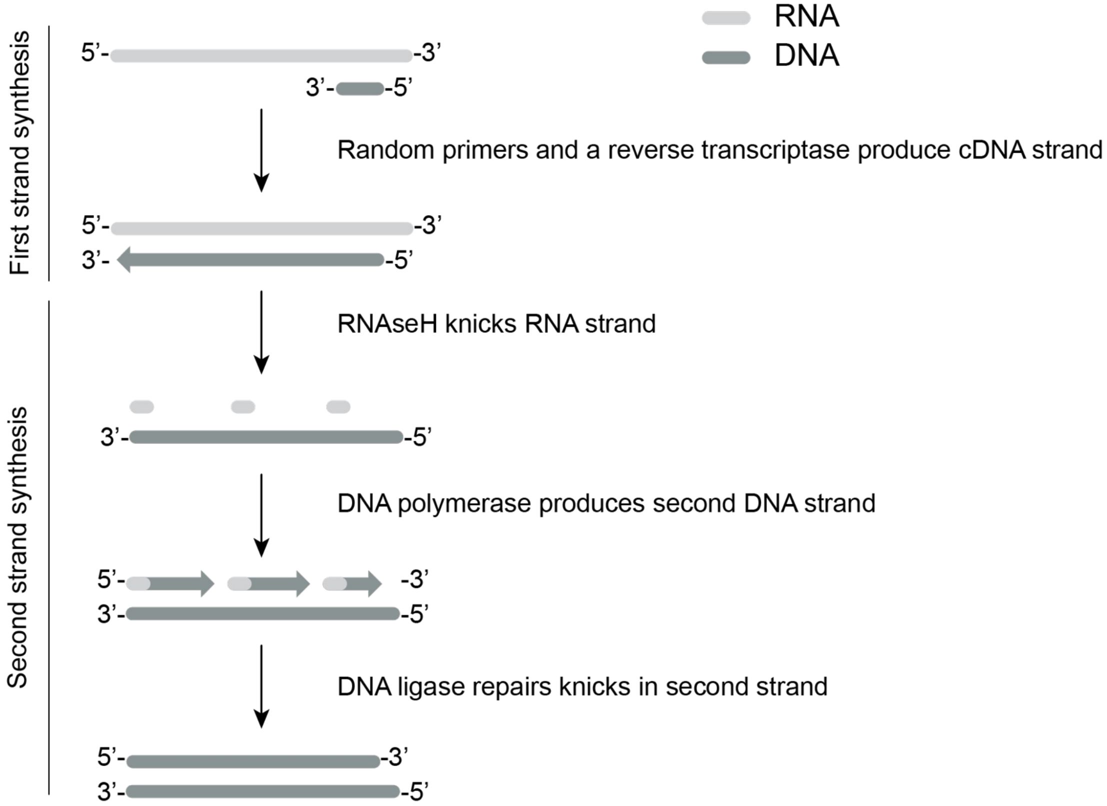
Red	Ladder
Blue	150-300 bp, mRNA fragmented for 15 minutes at 94°C
Green	200-500 bp mRNA fragmented for 10 minutes at 94°C
Cyan	400-1,000 bp mRNA fragmented for 5 minutes at 94°C

RNA library prep

STEP 3. dsDNA synthesis

1. First strand synthesis
2. Second strand synthesis
3. Purify dsDNA by

   SPRI bead clean-up (1.8x)

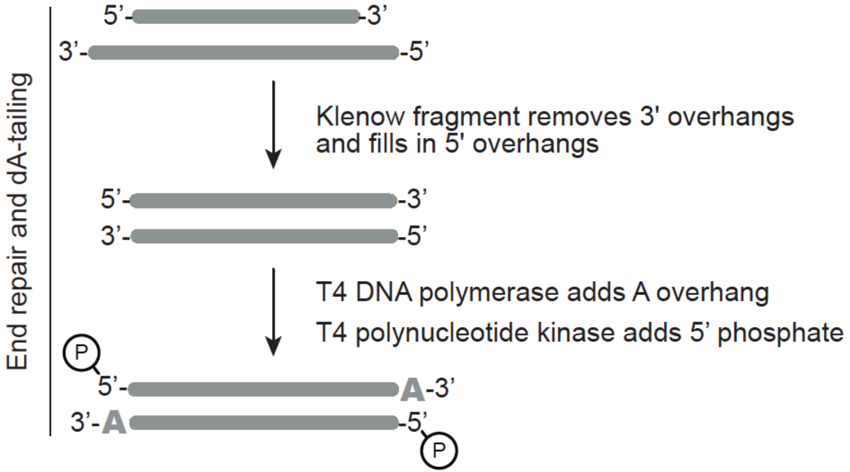


RNA library prep

STEP 4. End repair & dA-tailing

1. Remove overhangs to get blunt ends
2. Prep fragment to clone on adapters
    a. Add 5’ phosphate (T4 kinase)
    b. Add single 3’ A overhang to clone on adapters

(T4 DNA polymerase)



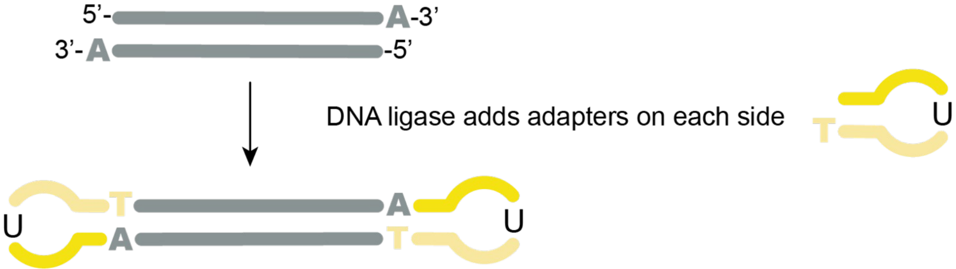
Note: After end-repair, move on to adapter ligation step ASAP

(5’ phosphate is fragile)

RNA library prep

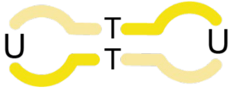
STEP 5. Adapter ligation

1. Adapters are added to allow sequencing

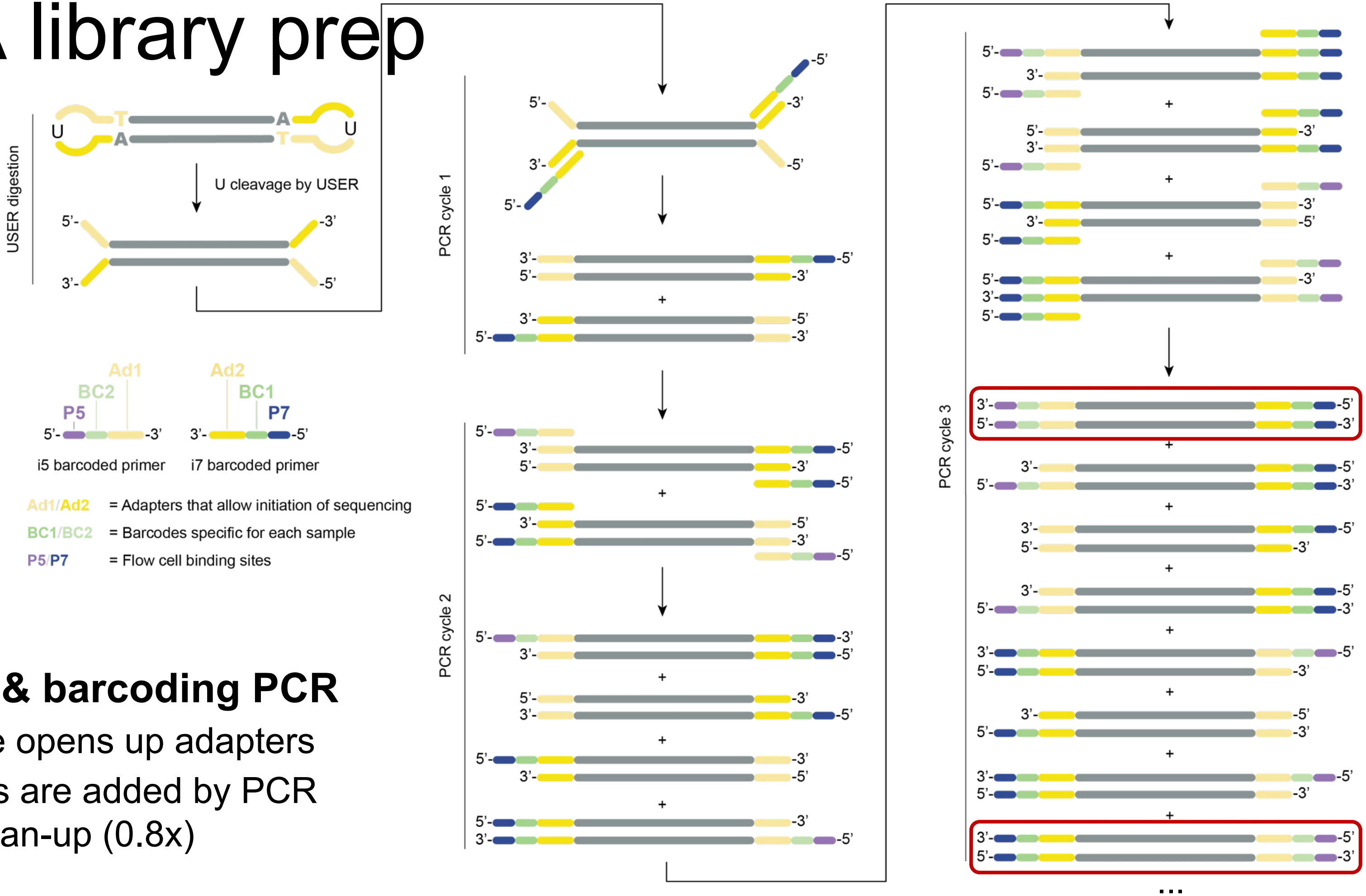


1. Remove adapter dimers by SPRI bead clean-up (0.9x)

Adapter dimers are our enemy #1 after this step (short fragments get preferentially sequenced)



RNA library prep



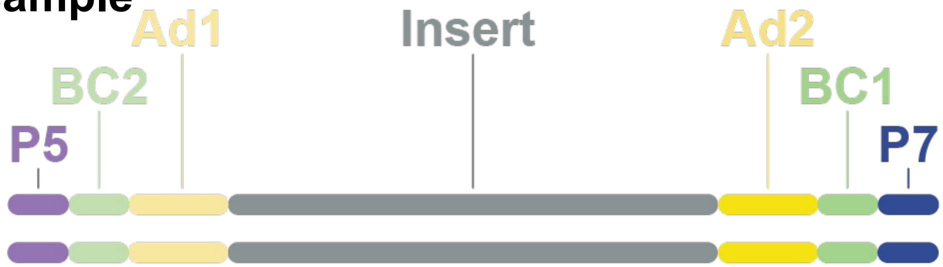
STEP 6.

USER digestion & barcoding PCR

1. USER enzyme opens up adapters
2. i5 and i7 oligos are added by PCR
3. SPRI bead clean-up (0.8x)

RNA library prep

Final library for one sample



Insert	= DNA fragment to be sequenced
Ad1/Ad2	= Adapters that allow initiation of sequencing
BC1/BC2	= Barcodes specific for each sample (one for each strand)
P5/P7	= Flow cell binding sites

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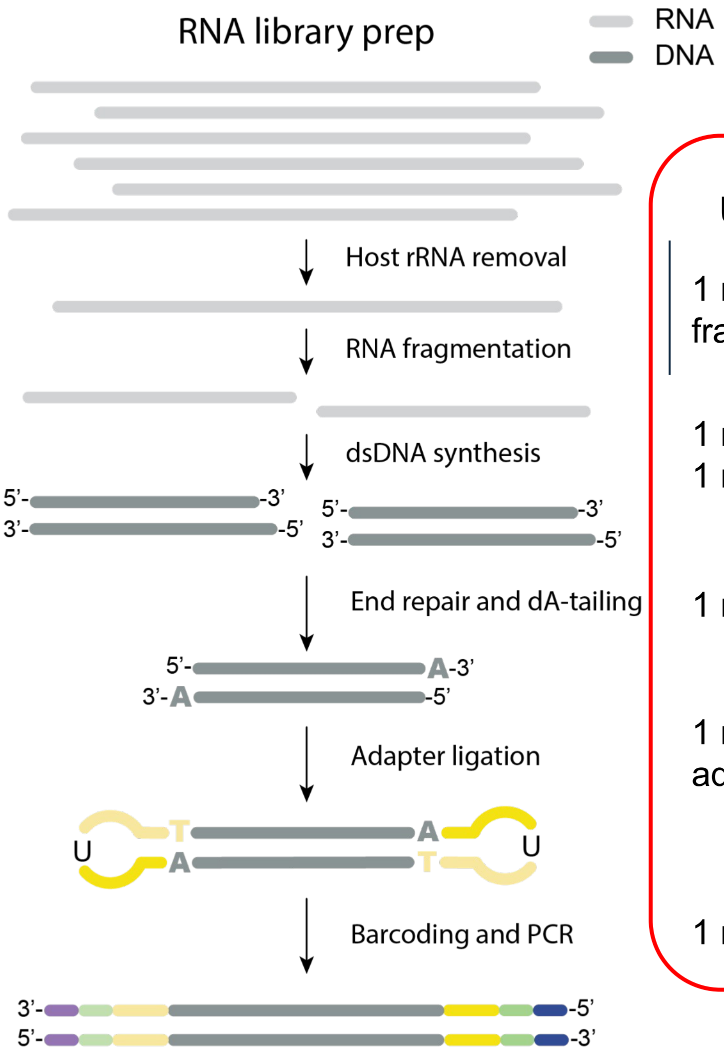
How to prepare a library in the lab?



RNA library kit options

NEBNext Ultra II RNA	Illumina TruSeq	Illumina Nextera
1 rxn, fragm. by Mg^{2+} & heat	1 rxn, fragm. by Mg^{2+} & heat	cDNA made separately
1 rxn for 1st str. synt. 1 rxn for 2nd str. synt.	1 rxn for 1st str. synt. 1 rxn for 2nd str. synt.	1 rxn, Transposome frag. and ligation of adapters
1 rxn	1 rxn	
1 rxn, U shaped adapters	1 rxn, Y shaped adapters	
1 rxn	1 rxn	1 rxn

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RNA library prep (NEBnext Ultra II RNA)

All in pre-PCR lab

A. RNA fragmentation, priming and host rRNA removal

1. Prepare mix

Reagents	**µl per reaction**
RNA sample (10ng – 100ng total)	4
First Strand Synthesis Reaction Buffer (5x)	4
Random Primers	1
*QIAseq FastSelect (1:100) rRNA	1
Total volume	**10**

2. Thermocycler:

Time depends on RNA quality!

A. 2-8 min at 94^{o}C: fragmentation
B. 2 min at 75^{o}C
C. 2 min at 70^{o}C
D. 2 min at 65^{o}C
E. 2 min at 60^{o}C
F. 2 min at 55^{o}C
G. 5 min at 37^{o}C
H. 5 min at 25^{o}C
I. Hold at 4^{o}C

Slow ramp for FastSelect to specifically anneal to host rRNA

RNA library prep

All in pre-PCR lab

B. First Strand Synthesis

1. Prepare mix

Reagents	**µl per reaction**
Fragmented and primed RNA	10
Nuclease-free water	8
NEBNext First Strand Synthesis Enzyme Mix	2
Total volume	**20**

2. Thermocycler:
   - A. 10 min at 25°C: random primer annealing
   - B. 15 min at 42°C: reverse transcription rxn
   - C. 15 min at 70°C: heat inactivation of enzymes
   - D. Hold at 4°C

RNA library prep

All in pre-PCR lab

C. Second Strand Synthesis

1. Prepare mix

Reagents	**µl per reaction**
First strand synthesized DNA	20
2nd Strand Synthesis Reaction Buffer (10x)	8
2nd Strand Synthesis Enzyme Mix	4
Nuclease-free water	48
Total volume	**80**

2. Thermocycler (HEATED LID OFF):
   A. 1 hr at 16°C: 2nd strand synthesis
   B. Hold at 4°C

3. SPRI bead cleanup (1.8x) to purify dsDNA

RNA library prep

All in pre-PCR lab

D. End prep of cDNA

1. Prepare mix

Reagents	**µl per reaction**
Purified dsDNA	50
Ultra II End Prep Reaction Buffer (8.6x)	7
Ultra II End Prep Enzyme Mix	3
Total volume	**60**

2. Thermocycler (heated lid at 105^{o}C):

   A. 30 min at 20^{o}C:
      - Klenow fragment creates blunt ends
      - T4 kinase adds 5' phosphate,
      - T4 DNA polymerase adds 3' A overhang

   A. 30 min at 65^{o}C: heat inactivation of enzymes

   B. Hold at 4^{o}C

3. Proceed immediately to next step, do NOT freeze.

RNA library prep

All in pre-PCR lab

E. Adapter ligation

1. Prepare mix

Reagents	**µl per reaction**
End prep reaction mixture	60
NEBNext Ultra II Ligation Master Mix	30
NEBNext Ligation Enhancer	1
1:10 Adapters (only add at the very end, do not pre-mix)	2.5
Total volume	**93.5**

2. Thermocycler (HEATED LID OFF):

   A. 15 min at 20°C

3. Proceed immediately to SPRI bead cleanup (0.9x)

RNA library prep

F. USER digestion and Barcoding PCR

Pre-PCR lab

1. Prepare mix

Reagents	**µl per reaction**
Purified, adapter-ligated dsDNA	15
USER Enzyme	3
NEBNext Ultra II Q5 Master Mix	25
5 µM i7 barcoded primer	5
5 µM i5 barcoded primer	5
Total volume	**53**

Post-PCR lab

2. Bring tubes/plate to POST-PCR lab

3. Thermocycler (Heated lid at 105^{o}C):

   A. 15 min at 37^{o}C: USER enzyme activity
   B. 30 sec at 98^{o}C
   C. 10 sec at 98^{o}C
   D. 75 sec at 65^{o}C
   E. 5 min at 65^{o}C
   F. Hold at 4^{o}C

   6 to 15 cycles (steps C–D)

4. SPRI bead cleanup to remove amplified dimers

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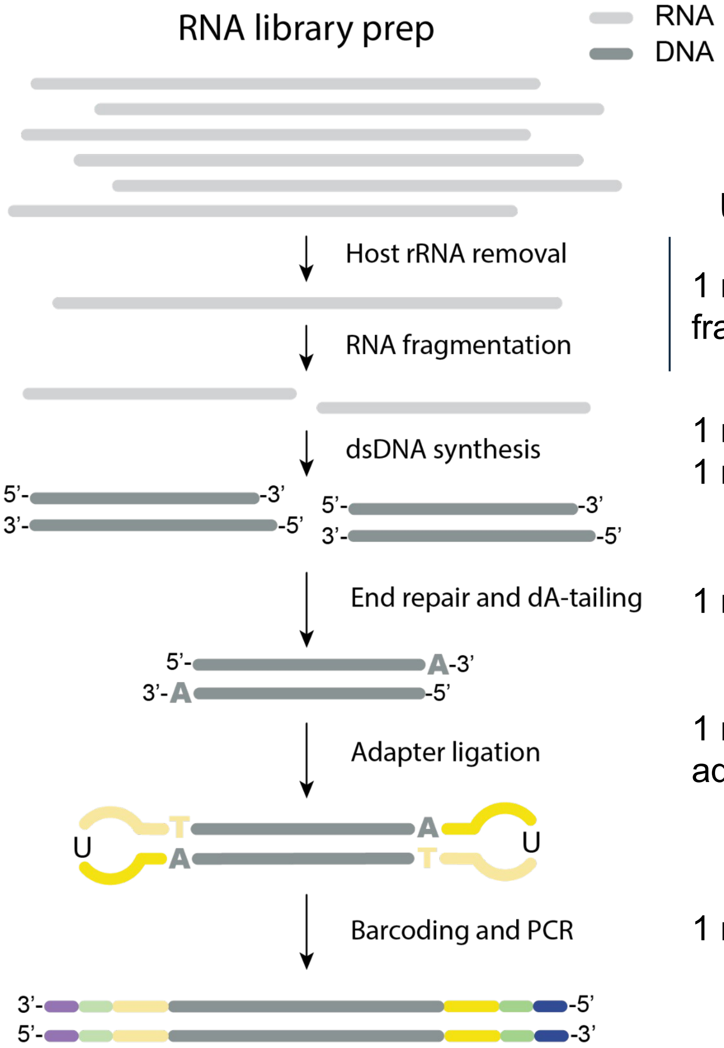
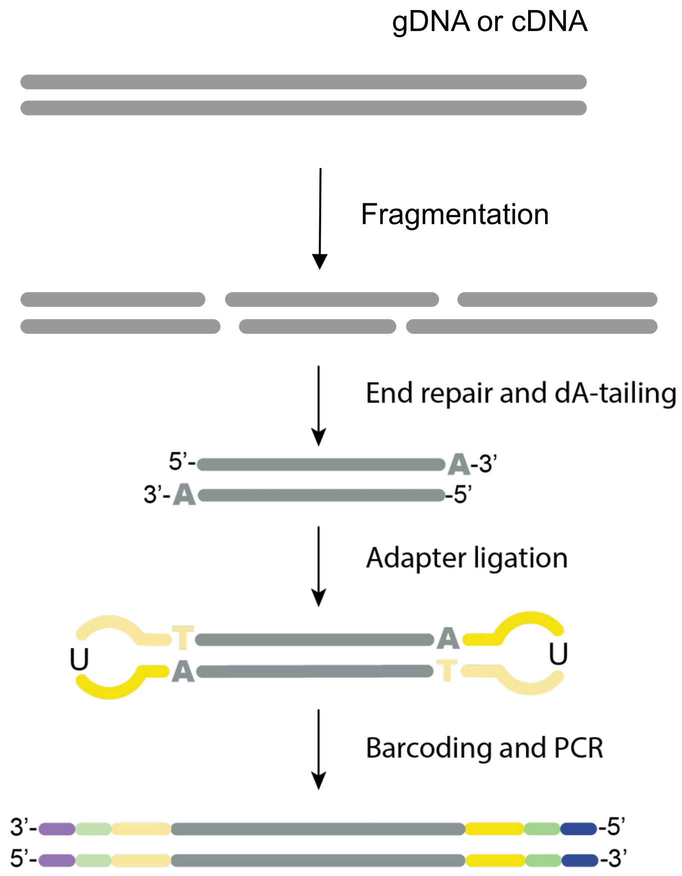


Step	NEBNext Ultra II RNA	Illumina TruSeq	Illumina Nextera
Host rRNA removal / RNA fragmentation	1 rxn, fragm. by Mg^{2+} & heat	1 rxn, fragm. by Mg^{2+} & heat	cDNA made separately
dsDNA synthesis	1 rxn for 1st str. synt. 1 rxn for 2nd str. synt.	1 rxn for 1st str. synt. 1 rxn for 2nd str. synt.	
End repair and dA-tailing	1 rxn	1 rxn	1 rxn, Transposome frag. and ligation of adapters
Adapter ligation	1 rxn, U shaped adapters	1 rxn, Y shaped adapters	
Barcoding and PCR	1 rxn	1 rxn	1 rxn

DNA library prep



DNA library kit options

NEBNext Ultra II FS DNA	**Illumina DNA Prep** WGS, cDNA, amplicons	**Illumina Nextera XT** (small genomes, cDNA, amplicons)
1 rxn, Fragm. enzyme + repair enzymes	1 rxn, Transposome frag. + ligation on bead	1 rxn, Transposome frag. + ligation in solution
1 rxn, U shaped adapters		
1 rxn	1 rxn	1 rxn

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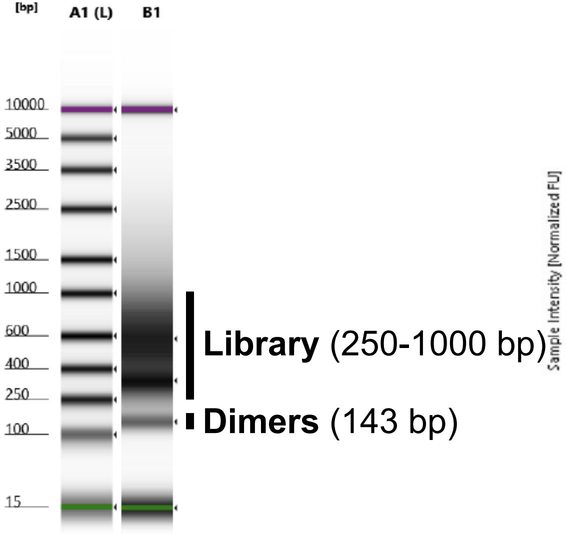
Library prep QC

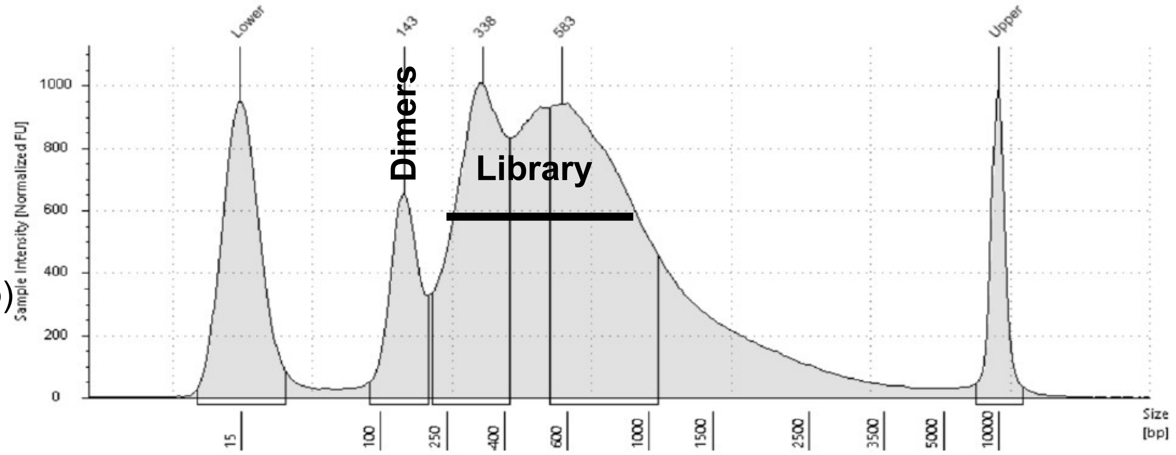
Post-PCR lab

Library QC: quantity & quality

1. Quantity (ng/µl): Qubit to determine concentration of individual libraries (0.5-10 ng/µl)

2. Quality (bp): Tapestation to determine (i) length of library and (ii) degree of dimers present





Library prep QC

Post-PCR lab

Library QC: quantity & quality

1. Quantity (ng/µl): Qubit to determine concentration of individual libraries (0.5-10 ng/µl)

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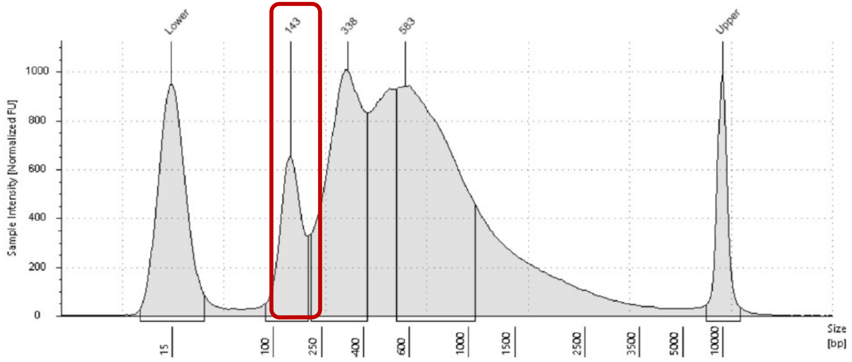
Size selection by SPRI beads
to remove dimers

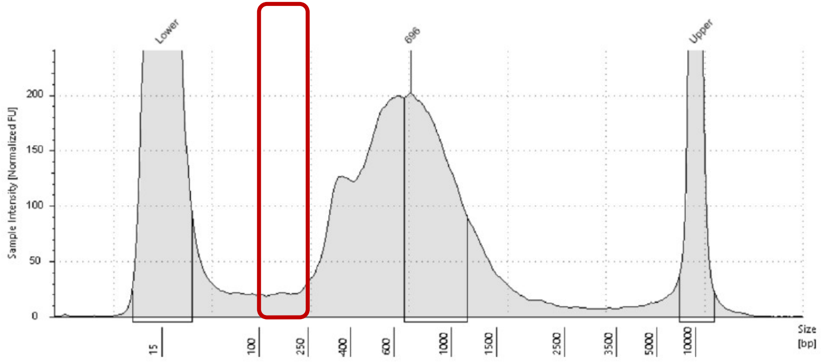
After PCR amplification



Adapter dimers

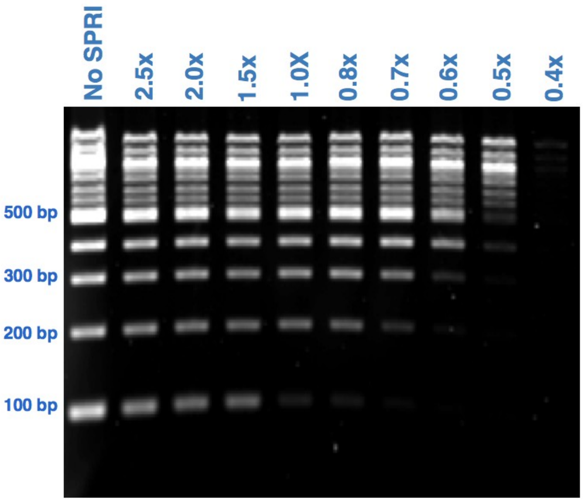
3 rounds
SPRI
(0.85x)

After SPRI bead clean-up



ALL DIMERS SHOULD BE REMOVED!

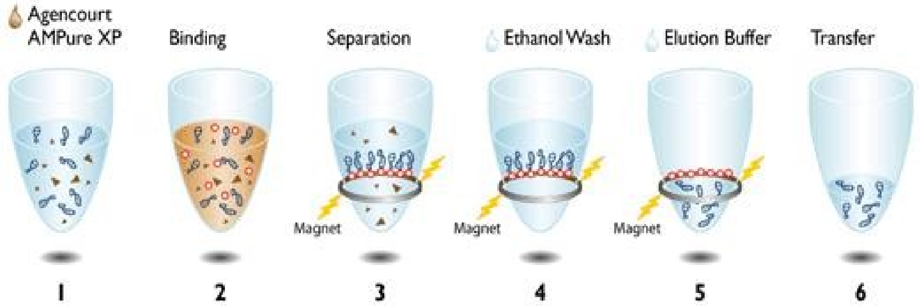
Library prep QC



AMPure XP SPRI select beads (paramagnetic)

or homemade SPRI beads (>10x cheaper)

- The longer a fragment, the better it will bind to the beads
- The fewer beads you add, the more short fragments you'll lose
- Perform size selection using SPRI beads to remove adapter dimers (143 bp)



1. PCR reaction 2. Binding of PCR amplicons to magnetic beads 3. Separation of PCR amplicons bound to magnetic beads from contaminants 4. Washing of PCR amplicons with Ethanol 5. Elution of PCR amplicons from the magnetic particles 6. Transfer away from the beads into a new plate

$$\text{SPRI ratio} = \frac{\text{Volume beads}}{\text{Volume reaction}}$$

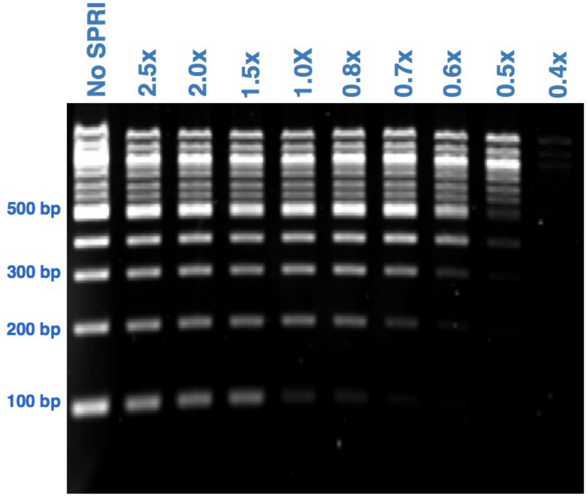
Example: for a 1.8x SPRI clean-up:

- Add 18 µl of SPRI beads to 10 µl of sample

Library prep QC

SPRI bead clean-ups during library prep

- Size selection using SPRI beads to remove adapter dimers
- Rule of thumb: dimers should be completely should disappear before

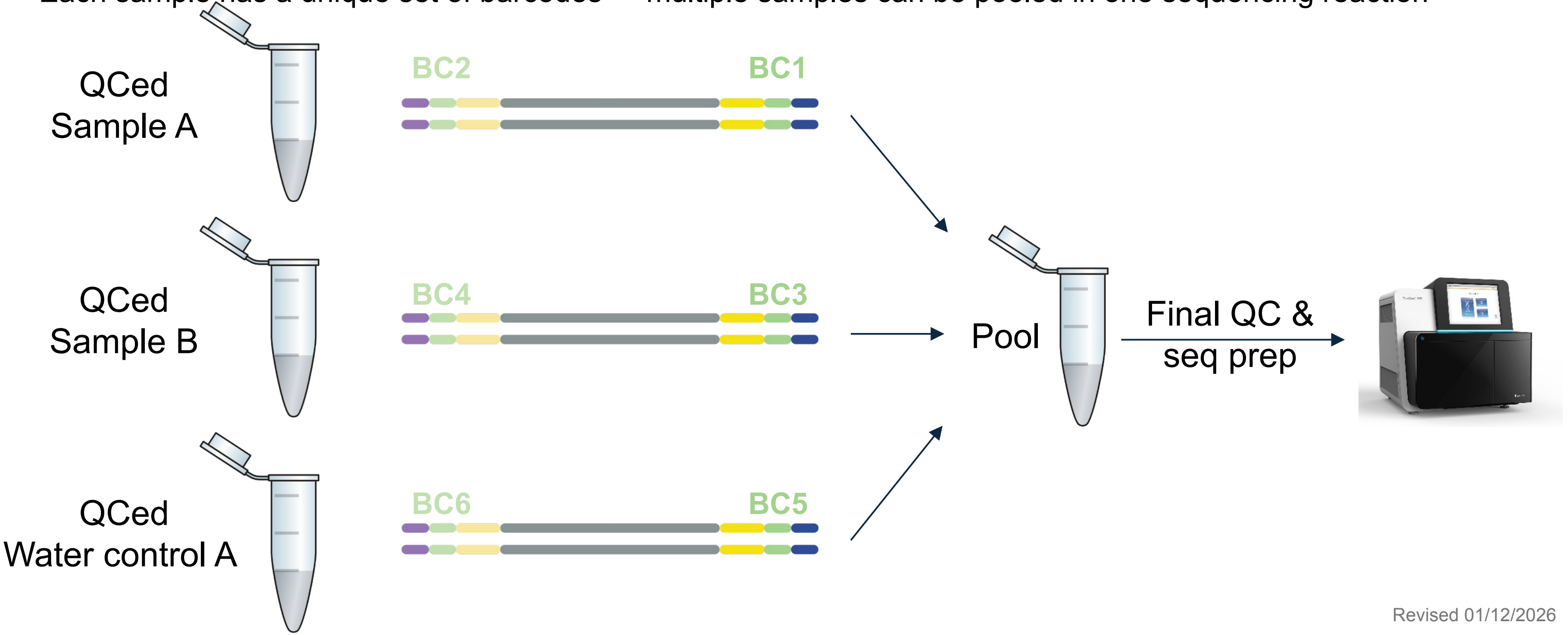


Step	Ideal RNA/library size	To remove	SPRI ratio
After DNAse I treatment	-	Reagents	
After second strand synth.	± 400 bp	Reagents	1.8x
After adapter ligation	± 464 bp	Adapter dimers (64 bp) & reagents	0.9x
After P7/P5 PCR	± 543 bp	P5-adapter dimer-P7 (143 bp) & reagents	0.8x
QC of library	± 543 bp	P5-Adapter dimer-P7 (143 bp)	0.8x

Library prep QC

Pooling samples for sequencing

- Each sample has a unique set of barcodes -> multiple samples can be pooled in one sequencing reaction



Library prep QC

Pooling samples together for sequencing

1. Qubit to determine concentration of individual libraries (expect library to be 0.5 to 10 ng/ul)
2. Pool barcoded libraries so that all are equally represented (except H_2O controls, only 1:5)

Qubit concentration			Pool		
Sample 1:	10 ng/ul	->	4.0 µl	->	40 ng
Sample 2:	5 ng/ul	->	8.0 µl	->	40 ng
Sample 3:	8 ng/ul	->	5.0 µl	->	40 ng
Sample 4:	15 ng/ul	->	2.6 µl	->	40 ng
H_2O control:	1 ng/ul	->	8.0 µl	->	8 ng

Total: 27.6 µl 168 ng

End goal is >10 µl of 2 nM pooled library (or 4 nM for MiSeq)

Library prep QC

Pick one way to calculate nM and stick with it!
- **Qubit concentration and library size**
- **Automatic calculation by Tapestation**

Pooling samples together for sequencing

3. After pooling, determine quantity and quality of pool

Qubit

$$\text{Concentration in nM} = \frac{\text{concentration in ng/µl}}{660\text{g/mol} \times \text{average library size in bp}} \times 10^6$$

Capillary or gel electrophoresis

Example: if concentration is 6.1 ng/µl, and average library size is 550 bp -> concentration is 16.8 nM

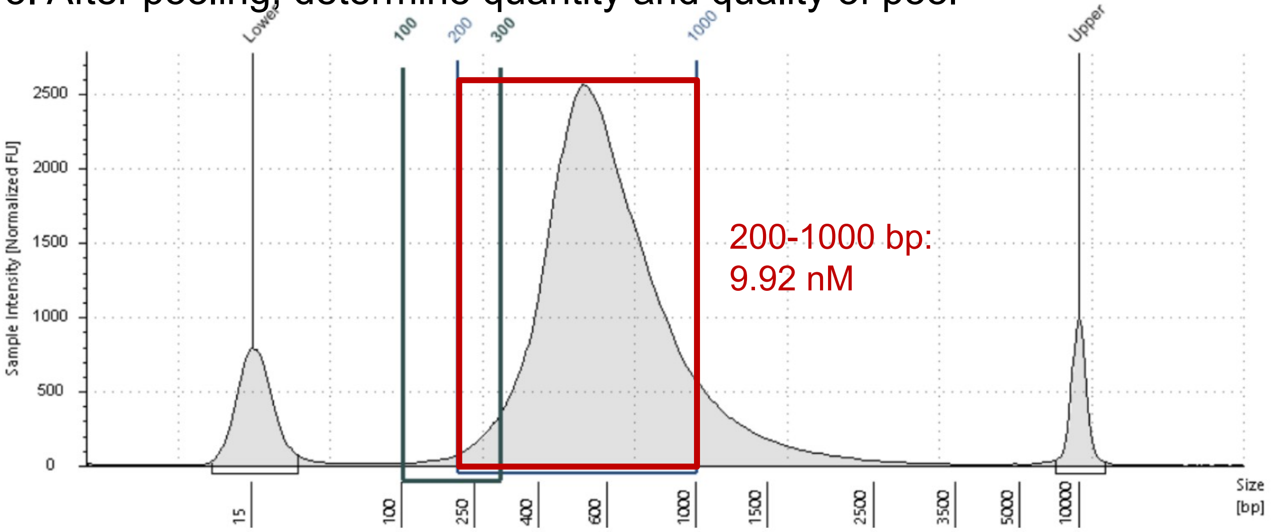
End goal is >10 µl of 2 nM pooled library

Library prep QC

Pick one way to calculate nM and stick with it!
- **Qubit concentration and library size**
- **Automatic calculation by Tapestation**

Pooling samples together for sequencing

3. After pooling, determine quantity and quality of pool



Or use Tapestation to calculate concentration of library in nM for fragments between 200-1000 bp long (see extra slides for a step-by-step guide on how to set that up)

Region Table

From [bp]	To [bp]	Average Size [bp]	Conc. [pg/µl]	Region Molarity [pmol/l]	% of Total	Region Comment	Color
100	300	249	114	783	2.95		
200	1000	577	3410	$\boxed{9920}$	88.29		

Library prep - preparing to load the sequencer

Dilute library to the correct loading concentration:

a. iSeq:
   i. dilute to 75 pM (or another optimized concentration),
   ii. Spike in 1-5% of PhiX at the same concentration

a. MiSeq:
   i. First denature library with sodium hydroxide (NaOH)
   ii. Then dilute to 10-12 pM (or another optimized concentration)
   iii. Spike in 1-5% of denatured PhiX to denatured and diluted library

See slide decks for iSeq or MiSeq loading for more details!

Next step:
iSeq or MiSeq loading